

Predictive Modelling of IPO Listing Gains

Domain: Capital Markets | **Tools:** SPSS, Excel | **Data:** 258 Mainboard IPOs, NSE/BSE (2016–2025)

Summary

Objective Test which pre-listing IPO signal — Grey Market Premium (GMP), subscription demand, or media sentiment — actually predicts listing-day gains, so investors can move from hype-driven decisions to evidence-based ones.

Concepts / Methods Used Multiple Linear Regression (3 model specifications), Pearson Correlation, Collinearity Diagnostics (Tolerance/VIF), Log Transformation (skew correction), Z-score Outlier Trimming with re-validation.

Key Output

- Full model: $R^2 = 0.638$ ($F = 88.93$, $p < .001$) — combined signals explain 64% of listing-gain variation
- GMP is the dominant predictor ($\beta = .705$, $p < .001$); subscription and sentiment lose significance in its presence
- Without GMP, subscription demand alone is significant ($R^2 = .419$, all three categories $p < .05$)
- Media sentiment is insignificant in every model tested ($p = .25$ to $.59$)
- Finding holds after outlier removal (258 → 239 IPOs; GMP remains significant at $p < .001$)

Business Actionables

- Use GMP as the primary screening signal for listing-gain potential
 - Use QIB/HNI subscription as a secondary confirmation check — a mismatch (high GMP with weak QIB demand) is a warning flag
 - Discount media buzz and hype — it has no measured predictive value
 - Evaluate on the full outcome distribution (range: -82% to $+253\%$), not the average gain, as downside risk is material
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Data & Proof of Analysis

Dataset: Self-compiled — GMP (averaged across sources), category-wise subscription, issue/listing price, media sentiment (coded from news and analyst coverage) **Sample:** 258 mainboard IPOs, reduced to 239 after outlier trimming **Dependent Variable:** Listing_gain (%) = $(\text{Listing Price} - \text{Issue Price}) / \text{Issue Price} \times 100$ **Independent Variables:** GMP_percentage, Log_retail, Log_HNI, Log_QIB, Net_sent

Evidence to include (from actual SPSS output file):

#	Output	Demonstrates
1	Descriptive Statistics table (N=258)	Data was loaded and profiled
2	Pearson Correlation matrix	Relationships between predictors tested
3	Model 1 — Summary, ANOVA, Coefficients (GMP dominant)	Core regression run; headline result
4	Model 2 — Coefficients (GMP excluded, subscriptions significant)	Isolation test confirming signal overlap
5	Model 3 — Coefficients (raw subscriptions + GMP)	Robustness across specifications
6	Collinearity Statistics (Tolerance/VIF)	Model validity check
7	Z-score output and trimmed-sample regression (N=239)	Outlier robustness check

Insert actual SPSS screenshots at the marked points in the final document.

Business Takeaway

When multiple signals overlap, testing them together — not in isolation — reveals which one actually carries information. Here, that signal is GMP.